Homework

Homework 1 — A sum and a threshold — 45 min, after S11.

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Prove by induction that $2 \leq u_n < 6$ for every $n \in \mathbb{N}$. Deduce the direction of variation of (u_n).
2. Let $v_n = u_n - 6$ for every $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in terms of n.
3. For every $n \in \mathbb{N}$, let $S_n = \sum_{k=0}^{n} u_k$. Express S_n in terms of n.
4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the previous integer does not work.
5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result by considering the mean of the first $n+1$ terms; no limit calculation is required.